

IR Transmission Research for Multi-Transmitter Systems

1 Introduction

While creating the NEC parallel transmitter, I encountered an interference problem when two transmitters are placed near each other. Therefore, I need to develop protection against this interference, such as a guiding tube or other isolation methods.

2 Signal Characteristics

My signal follows the NEC standard with a carrier frequency of **38 kHz**, and the wavelength of the transmission LED is **940 nm** [3], which places it in the **NIR (Near-Infrared)** region.

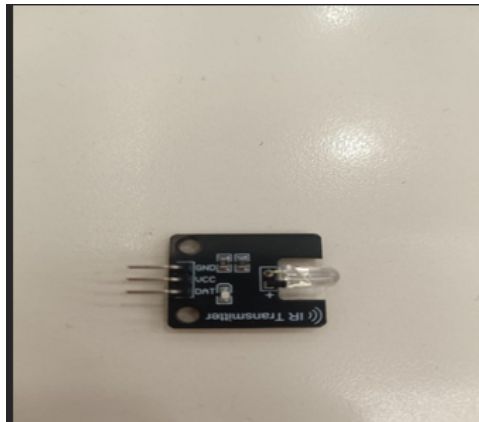


Figure 1: IR Transmitter Element

The spectral regions are illustrated in Figure 2.

3 Physical Isolation Using Reflective Materials

I searched for materials that reflect IR light to prevent interference between transmitters. Polished metals are commonly used for this purpose [4]. I used copper metal, and it works properly for insulation.

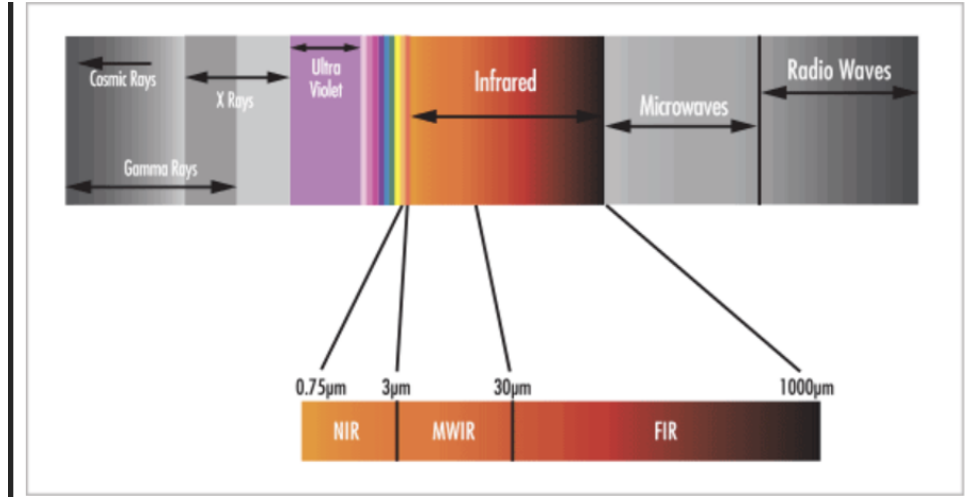


Figure 2: Infrared Spectral Regions [2]

4 System Expansion

I was asked to add more parallel transmitters. The maximum number of pins available for transmission is:

- 8 pins from PMOD A
- 8 pins from PMOD B
- 20 pins from Arduino

This gives a total of **36 pins** for transmission. I will use 24 transmitters: all Arduino pins and 4 PMOD A pins.

5 Frequency-Division Approach

Physical barriers are not feasible, so alternative methods for IR transmission must be implemented. The proposed approach is **carrier frequency shifting** from 38 kHz to higher frequencies.

5.1 Initial Plan

The initial plan was to shift from **38 kHz to 84 kHz** with **2 kHz steps**. However, both the transmitter and receiver are designed for 38 kHz operation.

5.2 Revised Frequency Plan

Based on research, IR receivers can detect signals near their center frequency. To ensure sufficient separation, a **10 kHz spacing** is required. Therefore, the new frequency range is from **38 kHz to 278 kHz**. This can be approached if we will use the TSAL6100 IR LED transmitter with different carrier frequency, we will do it using FPGA.

5.3 Receiver Characteristics

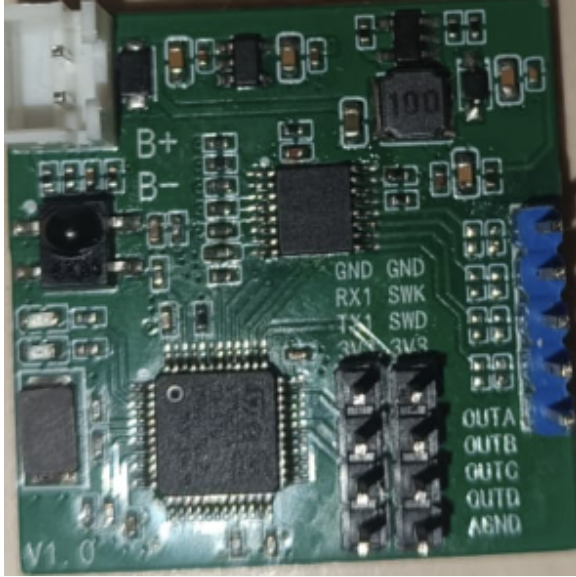
The main characteristics of the IR transmitter:

- Range: 1.3 m (at 5V and 38 kHz)
- Wavelength: 940 nm
- Carrier frequency: 38 kHz (fixed for standard modules)
- Also possible to receive from 35KHZ to 41KHZ

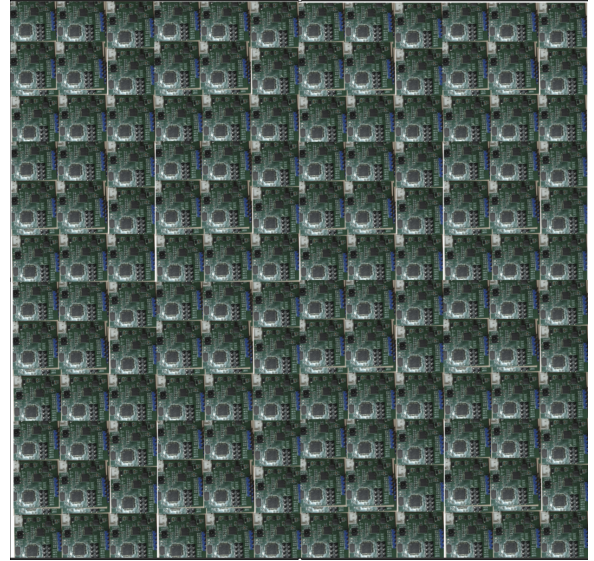
While more sophisticated approaches, such as channel reuse strategies for indoor infrared wireless communications, are documented in the literature [5], this work focuses on a specific hardware implementation.

6 Single-Transmitter, 144-Receiver Array Configuration

The experimental setup consists of a single infrared transmitter paired with an array of 144 individual IR receivers. Each receiver unit occupies a footprint of $2.5\text{ cm} \times 2.5\text{ cm}$ in a square form factor. The receivers are arranged in a contiguous 12×12 grid to form a larger square detection surface, resulting in a total active area of 900 cm^2 .



(a) Single IR receiver unit.



(b) Complete 12×12 receiver grid.

Figure 3: Infrared receiver configuration. (a) Individual $2.5\text{ cm} \times 2.5\text{ cm}$ receiver unit. (b) Full 144-receiver array spanning a total active area of 900 cm^2 .

6.1 Problems with the IR Transmission

The initial prototype employed the KY-005 infrared transmitter module [3]. However, this module proved inadequate for the required application due to its limited optical output power. The KY-005 is fundamentally a low-power, narrow-beam IR transmitter consisting solely of a 5 mm IR LED [7]. Its key specifications are summarized as follows:

- **Operating Voltage:** 5 V
- **Forward Current:** 30–60 mA
- **Power Consumption:** 90 mW
- **Operating Temperature:** -25°C to 80°C
- **Dimensions:** 18.5×15 mm
- **Compatibility:** Works in tandem with the KY-022 IR receiver module; suitable for Arduino, Teensy, Raspberry Pi, and ESP8266 platforms.

Preliminary testing confirmed that the KY-005 module is incapable of reliably illuminating the entire 12×12 receiver grid, which spans a total active area of 900 cm^2 and consists of 144 individual receiver units (see Figure 4). The module’s restricted radiant intensity and narrow emission cone result in significant signal attenuation at the grid’s periphery, rendering it unsuitable for wide-area coverage.

6.1.1 Proposed Solution: High-Power IR LED Array

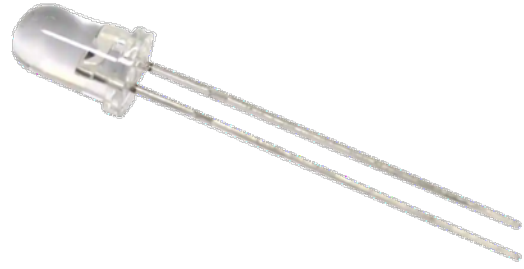
To overcome these limitations, the KY-005 module is to be replaced with a discrete high-power infrared LED driven by a dedicated external circuit. The selected IR LED component (sourced from Lite-On [8]) offers a substantially enhanced performance envelope, as detailed by the following electrical and optical characteristics:

- **Wavelength Range:** 850–940 nm
- **Beam Angle / Viewing Angle:** 16° to 150°
- **Forward Voltage (V_F):** 0.8–2.5 V
- **Reverse Voltage (V_R):** 4–40 V
- **Forward Current (I_F):** 1 mA to 1 A (pulsed)
- **Power Rating:** 100 mW to 2.5 W
- **Radiant Intensity:** 2.165–400 mW/sr
- **Rise/Fall Time:** 7.5–40 ns
- **Peak Soldering Temperature:** $+260^{\circ}\text{C}$

The increased forward current capacity (up to 1 A pulsed) and higher radiant intensity ensure sufficient optical flux to saturate all 144 receivers across the 900 cm^2 area. The accompanying driver circuit is designed to modulate the LED at the required 38 kHz carrier frequency while supplying the necessary peak current without exceeding the component’s thermal limits.



(a) Single IR receiver unit (2.5×2.5 cm).



(b) Complete 12×12 receiver grid.

Figure 4: Infrared receiver configuration. (a) Individual receiver unit. (b) Full 144-receiver array covering a total active area of 900 cm^2 .

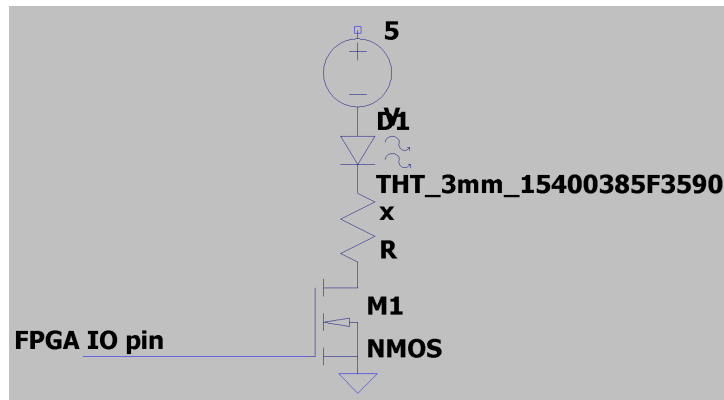


Figure 5: Schematic of the high-current driver circuit for the high-power IR LED.

References

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